

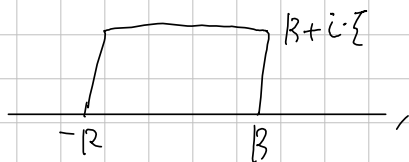
✓ Evaluate

$$\int_{-\infty}^{\infty} e^{-\pi x^2} \cdot e^{-2\pi i x \zeta} dx = e^{-\pi \zeta^2} \quad (\zeta > 0)$$

Proof. Let  $f(z) = e^{-\pi z^2}$ . Then, it is entire.

Consider four contour:  $C_1: t \quad (-R \leq t \leq R)$ ,  
 $C_2: R + it \quad (0 \leq t \leq \zeta)$   
 $C_3: t + i\zeta \quad (-R \leq t \leq R)$   
 $C_4: -R + it \quad (0 \leq t \leq \zeta)$

Then, the composition  $C = C_1 \cdot C_2 \cdot (-C_3) \cdot (-C_4)$  is described as



and, by the Cauchy's theorem,  $\int_C f = 0$ . Now,

$$\int_{C_1} f = \int_{-R}^R e^{-\pi x^2} dx,$$

$$\int_{C_2} f = \int_0^{\zeta} e^{-\pi(R+it)^2} \cdot i dt = i \int_0^{\zeta} e^{-\pi(R^2 - t^2 + 2iRt)} \cdot dt$$

$$\text{and so } \left| \int_{C_2} f \right| = \left| \int_0^{\zeta} e^{\pi \cdot (t^2 - R^2)} dt \right| \leq \left| \int_0^{\zeta} e^{\pi t^2} dt \right| \cdot e^{-\pi R^2} \rightarrow 0 \text{ as } R \rightarrow \infty.$$

Similarly  $\left| \int_{C_4} f \right| \rightarrow 0$  as  $R \rightarrow \infty$ . And,

$$\begin{aligned} \int_{C_3} f &= \int_{-R}^R e^{-\pi(t+i\zeta)^2} dt = \int_{-R}^R e^{-\pi(t^2 - \zeta^2 + 2it\zeta)} dt \\ &= e^{\pi \zeta^2} \cdot \int_{-R}^R e^{-\pi t^2} \cdot e^{-2\pi it\zeta} dt \end{aligned}$$

Thus, as  $R \rightarrow \infty$ ,

$$0 = \int_{-\infty}^{\infty} e^{-\pi x^2} dx - e^{\pi \zeta^2} \int_{-\infty}^{\infty} e^{-\pi t^2} \cdot e^{-2\pi it\zeta} dt. \quad \square$$

For each  $n=2,3,\dots$  evaluate

$$\int_0^{\infty} \frac{1}{1+x^n} dx.$$



Proof. Let  $f(z) = \frac{1}{1+z^n}$ . Take a Contour  $C$  as a Composition of:

$$C_R: t \mapsto t \quad (0 \leq t \leq R), \quad C_L: t \mapsto R \cdot e^{it/n} \quad (0 \leq t \leq 2\pi),$$

$$C_S: t \mapsto t \cdot e^{2\pi i/n} \quad (0 \leq t \leq R), \text{ that is, } C = C_R * C_L * (-C_S).$$

Inside  $C$ ,  $1+z^n$  has only one zero,  $\exp(i\pi/n)$ , so  $f$  has only one singularity inside  $C$ , the simple pole  $\exp(i\pi/n)$ , by the Residue theorem,

$$\begin{aligned} \int_C f &= 2\pi i \cdot \text{Res}(f, \exp(i\pi/n)) = \lim_{z \rightarrow \exp(i\pi/n)} (z - \exp(i\pi/n)) \cdot \frac{1}{1+z^n} \cdot 2\pi i \\ &= \frac{2\pi i}{n \cdot \exp(i\pi/n)^{n-1}} = -\frac{\exp(i\pi/n)}{n} \cdot 2\pi i \end{aligned}$$

Meanwhile,

$$\begin{aligned} \left| \int_{C_L} f \right| &= \left| \int_0^{2\pi} \frac{1}{1+R^n \cdot e^{it}} \cdot \frac{Ri}{n} \cdot e^{it/n} dt \right| \\ &= \int_0^{2\pi} \frac{R/n}{|1+R^n \cdot e^{it}|} dt \leq \int_0^{2\pi} \frac{R/n}{|R^n - 1|} dt \rightarrow 0 \text{ as } R \rightarrow \infty. \end{aligned}$$

and

$$\begin{aligned} \int_{C_S} f &= \int_0^R \frac{e^{2\pi i/n}}{1+t^n e^{2\pi i}} dt = \frac{e^{2\pi i/n}}{n} \int_0^R \frac{1}{1+t^n} dt. \text{ So that} \\ -\frac{2\pi i}{n} \cdot \exp(i\pi/n) &= \lim_{R \rightarrow \infty} \int_C f = \int_0^{\infty} \frac{1}{1+x^n} dx - \int_0^{\infty} \frac{1}{1+x^n} dx \cdot e^{2\pi i/n} \\ &= (1 - e^{2\pi i/n}) \cdot \int_0^{\infty} \frac{1}{1+x^n} dx \end{aligned}$$

$$\Rightarrow \int_0^{\infty} \frac{1}{1+x^n} dx = -\frac{2\pi i}{n} \cdot \exp(i\pi/n) / (1 - e^{2\pi i/n}).$$

□

Evaluate  $\int_0^\infty \frac{x^2}{1+x^6} dx$

Proof. Let  $C_R$  be a closed curve of composition  $C_R = C_x * C_r * (-C_s)$  where  $C_x: t \mapsto t$  ( $0 \leq t \leq R$ ),  $C_r: t \mapsto R \cdot e^{i\pi t/6}$  ( $0 \leq t \leq 2\pi$ ),  $C_s: t \mapsto t \cdot e^{i\pi/3}$  ( $0 \leq t \leq R$ )

Inside the curve  $C$ ,  $f(z) = z^2/(1+z^6)$  has only one singularity at  $z_0 = \exp(i\pi/6)$ . By the Residue theorem

$$\int_{C_R} f = 2\pi i \cdot \text{Res}(f, z_0) = 2\pi i \cdot \lim_{z \rightarrow z_0} (z - z_0) \cdot \frac{z^2}{1+z^6} = z_0^2 \cdot \frac{2\pi i}{6z_0^5} = \frac{2\pi i}{6z_0^3} = \frac{\pi}{3}$$

Meanwhile, for  $R > 1$ ,

$$\left| \int_{C_R} f \right| \leq \int_0^{2\pi} \left| \frac{R^2 e^{i\pi t/3}}{1 + R^6 e^{i\pi t}} \right| \left| \frac{\pi i \cdot R \cdot e^{i\pi t/6}}{6} \right| dt$$

$$\leq \int_0^{2\pi} \frac{R^3}{|R^6 - 1|} \cdot \frac{\pi}{6} dt \rightarrow 0 \text{ as } R \rightarrow \infty$$

$$\int_{C_s} f = \int_0^R \frac{x^2 \cdot e^{2\pi i/3}}{1+x^6} \cdot e^{\pi i/3} \cdot dt = - \int_0^R \frac{x^2}{1+x^6} dx$$

Consequently,

$$\frac{1}{3\pi} = \lim_{R \rightarrow \infty} \int_{C_R} f = 2 - \int_0^\infty \frac{x^2}{1+x^6} dx, \text{ so } \frac{\pi}{6} \text{ is the desired. } \square$$

Evaluate  $\int_0^{\infty} \sin(x^2) dx$ . Is it equals to  $\int_0^{\infty} \cos(x^2) dx$

Evaluate  $\int_0^{\infty} \frac{\sin x}{x} dx$ , using indented semicircle.

√ Evaluate  $\int_0^{\infty} \frac{\log x}{1+x^4} dx$